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Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
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Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Herlach, D. M. (Inst. fur Raumsimulation) Holland Mirtz, D. (Inst. fur Raumsimulation) Gangopadhyay, A. K. (Washington University) Hyers, R. W. (University of Massachussets)		
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Task Description:	Undercooling measurements and structural and physical property studies are made as a function of composition in Ti/Zr-TM liquids that form quasicrystals and related crystal phases. Quantitative measurements of the melt viscosity and specific heats are made as a function of alloy composition near the ideal stoichiometry of the icosahedrally-ordered phases in select Ti/Zr-alloys. These are needed to study the onset of icosahedral order in undercooled liquids and to accurately model nucleation as a function of alloy composition. They are particularly important for probing the role of oxygen in the stabilization of icosahedral order in the Ti-TM-Si-O alloys. The high liquidus temperatures and strong reactivities of the Ti/Zr-TM and Ti-TM-Si-O alloys prohibit accurate measurements of these quantities without containerless solidification. Ground-based investigations are, therefore, being made with the electrostatic levitator (ESL) located at Marshall Space Flight Center, to discern the relative roles of oxygen and diffusion on the nucleation process, make initial measurements of the physical properties, and determine the phase diagrams of the alloys studied. These data will allow more quantitative measurements of these physical properties and benchmark nucleation measurements to be made using the European Space Agency MSL-EML facility on the International Space Station. The ISS experiments will also allow the first investigations of the role of diffusion on nucleation in the absence of convection. The nucleation data will be compared with the predictions from a new model developed under partial support by this contract. This research focuses on studies of nucleation processes when diffusion effects are significant, on the formation and structures of highly ordered noncrystallographic phases, and on the connections between the atomic structures of these phases and the structures and thermophysical properties of liquids. The products of this research will lead to an improved understanding of nucleation and growth processes and will improve our ability to optimize desired microstructures for modern technological materials, both on earth and in reduced gravity environments. The new information gained will also provide valuable insight into the nucleation and growth mechanism of complex periodic and nonperiodic phases (i.e. quasicrystals), and how the liquid structure influences the microstructural development. Nucleation is the initial step in most first order phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. An improved ability to control nucleation lies at the heart of the development of novel phase microstructures for advanced materials applications. The studies being carried out under the support of this NASA contract will expand our understanding of nucleation when the compositions of the initial and final phases are different, and when diffusion kinetics due to partitioning are competitive with interfacial attachment kinetics. They will lead to an improved understanding of the structures of undercooled liquids and polytetrahedral solid phases, of the growth of complex periodic and ordered nonperiodic phases, and of the mechanisms by which oxygen stabilizes icosahedral order in Ti-TM-Si-O alloys. The results of these studies will impact a number of technological areas in materials science, yielding an improved ability to develop new materials and phase microstructures and to better tune them to desired applications. The results of the ISS and ground-based research will be important for developing new materials for application to space exploration and materials processing in low gravity environments. In a low gravity environment, diffusion processes can dominate over the convective processes normally encountered with terrestrial processing of materials. It is essential, then, to develop a better understanding of microstructural development in such an environment; our studies of diffusion-influenced nucleation are a critical step toward that goal. As we have already demonstrated in our ground-based studies leading to the flight experiments, new techniques and technologies will also emerge that will impact broad areas of science and technology. Recently, working with representatives from NASA Marshall Space Flight Center, Huntsville, AL, and the Advanced Photon Source, we have developed a new technique (BESL) to obtain rapid structural information at high temperature on electrostatically levitated samples. This has already led to the verification of a half-century old hypothesis about nucleation from a metallic liquid. We have also demonstrated that such short-range order is common in metallic liquids and likely plays an important role in glass formation. This is important, since metallic glasses have been demonstrated to frequently have superior mechanical properties over crystalline materials. We have recently demonstrated that this technique can be applied to the rapid determination of the phase diagrams of high temperature materials, of central importance to the development of new materials for advanced propulsion for aerodynamic and space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
	Over the past year, under partial support by this contract, our ground-based research has continued studies of the equilibrium and metastable phase diagram studies on the alloys identified for ISS study, developed computer modeling methods and analyzed nucleation rate data from supercooled liquid metals, and correlated the nucleation barrier to crystal phases with growing short-range order in the liquid phase with supercooling. In addition, we have further analyzed data taken on a range of metallic liquids with BESL (Beamline Electrostatic Levitation), a facility invented under the support of this contract. A manuscript describing BESL was recently submitted to the Review of Scientific Instruments. With BESL, we have measured structural data on liquids to a deeper degree of supercooling than ever before, obtaining new fundamental experimental information that is important for developing a better understanding of stable and metastable liquids and the phase transitions that occur in them. Our developed ability to simultaneously measure structure and physical properties of liquids and their influence on crystallization processes and microstructural refinement is unprecedented and of tremendous interest to a broad scientific community, evidenced by the high profile publications and frequent invited presentations. The key results obtained since the last task progress report given in May, 2005 are listed below. Aspects of some of the research activities compliment NSF sponsored research in Kelton's group at Washington University, supported by NSF DMR DMR 03-07410. 1. The first results from BESL studies of supercooled TiZrNi liquids (Phys. Rev. Lett., 90, 195504/1-4 (2003)) provided the first demonstration of a half-century-old hypothesis made by Frank. A measured growing icosahedral order in the supercooled liquid caused the nucleation of a metastable icosahedral quasicrystal instead of the thermodynamically stable C14 Laves phase, linking the structure of the supercooled liquid with the phase selection during crystallization. During the past year, we have demonstrated that this can have a profound influence on metallic glass formation. It was observed earlier by us (L. Q. Xing, Y. T. Shen, K. F. Kelton, Appl. Phys. Lett., 81, 3371-3373 (2002)) that the substitution of small amounts of Ti for Zr in a Zr62Cu20Ni8Al10 alloy led to bulk metallic glass formation and the crystallization to the icosahedral quasicrystal. This suggested that the addition of Ti induces icosahedral ordering in the liquid, increasing the nucleation barrier for crystal phases that might form during cooling. From BESL measurements of the liquids in the equilibrium and supercooled state, we demonstrated that all of the liquids develop icosahedral order.

Task Progress:	With increasing Ti concentration, however, it develops at higher temperatures, explaining the easier glass formation. This is the first time that the connection between short range order of the liquid and glass formation has been demonstrated so directly. This work is being prepared for publication. 2. Using BESL, we have studied the short-range order in equilibrium and supercooled liquids of early (Ti) and late (Ni) transition metals. While theoretical calculations give reasonable predictions of the peak locations in the measured x-ray structure factors, S(q), for Ni, they fail to predict the developing shoulder on the high-q side of the second peak in Ni, which is a signature of developing icosahedral order. The profile of the second peak in S(q) is different in Ti, consistent with a distorted icosahedral order in the liquid. Although such distortions, resulting from a competition between structural (maximum number of bonds) and electronic (Jahn-Teller effect) factors, have been predicted in atomic cluster studies, our results are the first direct experimental observations of such distortions in a liquid. This is also the first time that the degree of distortion has been correlated with atomic interactions. It signals a competition between a tendency for atomic close packing in the nearly filled d-band late transition metals, which favors local icosahedral order, and the importance of the angular dependence of the d-band bonding in the early transition metals. Distorted icosahedral (or decahedral) short-range order is also expected in other early transition metal liquids. While the nucleation barrier for simple crystal phases is expected to be little affected by the degree of perfection of ISRO in alloy liquids, the latter could have a dramatic effect on the nucleation of ordered phases with icosahedral short-range order, as occurs in icosahedral quasicrystals and their related crystal approximant phases. This work was published in summer 2004 (Phys. Rev. Lett., 93, 037802 (2004). 3. Changes in the coordination number for liquid silicon upon supercooling, signaling an underlying liquid-liquid phase transition, have been widely predicted, based on molecular dynamics studies using a Stillinger-Weber potential. Previous X-ray and neutron measurements, however, have produced conflicting reports. Some studies have found an increase in the first shell coordination as the temperature decreases in the supercooled regime, while others have reported increases in the coordination number with decreasing temperature. We recently employed the BESL technique, coupled with high energy x-ray diffraction (125 keV), and rapid data acquisition (100ms collection times) using an area detector, to obtain high quality structural data more deeply into the supercooled regime than has been possible before. No change in coordination number was observed in this temperature region, calling into question previous experimental claims of structural evidence for the existence of a liquid-liquid phase transition. Instead, we found a gradual change in the next nearest neighbor shell, that is consistent with a continuous change from the metallic liquid structure toward that of the diamond cubic semiconductor. This work has been submitted to Physical Review Letters. 4. A key goal of this project is to determine the role of oxygen in the development of icosahedral short range order in Ti-Fe-Si-O alloys. To address this, we are working in collaboration with D. Herlach and D. Holland-Moritz, DLR, Cologne, Germany, to make in-situ x-ray diffraction studies of electromagnetically and electrostatically levitated Ti-Fe-Si-O liquids, following the phase formation sequence during solidification of the undercooled liquids. Using BESL, we have measured the complete diffraction patterns (up to q = 17Å-1) of these alloys at a rate of 30 patterns/sec during cooling and heating. Based on our joint experiments, the 1/1 approximant phase is the primary crystallizing phase at deep undercoolings, which is in contrast with binary (TiFe) and ternary (TiFeSi) alloys, where simple crystal structures form. Measurements of Ti-Fe-Si-O alloy taken upon heating, however, indicate that the 1/1 is metastable at high temperatures. It forms, therefore, because the nucleation barrier is less than for the crystal phase, consistent with the liquid having significant icosahedral short range order. A manuscript presenting and discussing these results will be submitted by early April, 2005. 5. In last year's report, we reported the observation of a maximum in the specific heats of undercooled Ti-Zr-Ni liquids at temperatures far above possible glass transition temperatures. Measurements made recently by us demonstrate that it is not an artifact caused by a changing emissivity. In addition to the specific heat data, two experimental campaigns at MSFC ESL yielded ~ 500 GB of data on the density, surface-tension, and viscosity of quasicrystal-forming and glass-forming alloys. Significant improvements in numerical methods were made that improved the precision of all three measurements. Correlated with the specific heat maximum, these data reveal an inflection in the density, a rising viscosity, and a sudden decrease in the surface tension, strongly indicating the presence of a transition in the liquid. We have shown that this is not associated with crystallization, compositional phase separation, or an electronic transition in the liquid. It signals either a liquid-liquid transition, or a dynamical crossover effect, in a highly fluid liquid but with a rapid decrease in configurational entropy. Our BESL x-ray diffraction measurements indicate that it is correlated with growing icosahedral order in the liquid. Within the past year, we have observed similar maxima in many liquids that form bulk metallic glasses. This work is being prepared for submission to Physical Review Letters.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
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Articles in Peer-reviewed Journals	Lee, G. W.; Gangopadhyay, A. K.; Kelton, K. F.; Hyers, R. W.; Rathz, T. J.; Rogers, J. R.; Robinson, D. S. "Difference in Icosahedral Short-Range Order in Early and Late Transition Metal Liquids" Physical Review Letters, vol. 93, 037802-1 , Jul-2004
Articles in Peer-reviewed Journals	Kelton, K. F. "Crystallization of liquids and glasses to quasicrystals" Journal of Non-Crystalline Solids, vol. 334&335, pp. 253-258 , Mar-2004
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Presentation	Kelton, K. F. "Getting the Hot Structures" Plenary Lecture, 2004 Denver X-ray Conference, Steamboat Springs, CO Aug-2004
Presentation	Kelton, K. F. "Amorphous Metal Formation and Crystallization - Couopled Processes in Crystal Nucleation" Keynote Lecture,11th International Symposium on Metastable, Mechanically Alloyed and Nanocrystalline Materials 2004 (ISMANAM-2004), Sendai, Japan Aug-2004
Presentation	Kelton, K. F. "Local Icosahedral Order in Undercooled Metallic Liquids and the Influence on the Nucleation Barrier" 12th International Conference on Liquid and Amorphous Metals, Metz, France Jul-2004
Presentation	Kelton, K. F. "Amorphous Metal Formation and Crystallization - Coupled Processes in Crystal Nucleation" Invited Seminar, Oak Ridge National Laboratory, Oct-2004
Presentation	Kelton, K. F. "Local Icosahedral Order in Undercooled Metallic Liquids and the Influence on the Nucleation Barrier" Invited seminar, Department of Materials Science and Engineering, University of Tennessee, Knoxville, TN Oct-2004
Presentation	Kelton, K. F. "Influence of Local Structure and Alloy Composition on Glass Formation and Stability" Invited presentation, “Hume Rothery Symposium: The Science of Complex Alloys,“ 134th Annual TMS Meeting. Feb-2005
Presentation	Kelton, K. F. "Icosahedral Order in Undercooled Metallic Liquids and the Influence on the Nucleation Barrier" Invited presentation, “Neutron Diffraction Characterization of Mechanical Behavior: Deformation II,“ 134th Annual TMS Meeting. Feb-2005
Presentation	Kelton, K. F. "Coupled Processes in Nucleation" Invited presentation, “Frontiers in Solidification Science: Nucleation,“ 134th Annual TMS Meeting. Feb-2005
Presentation	Bradshaw, R. C., Hyers, R. W., Rogers, J. R., Rathz, T. J., Lee, G. W., Gangopadhyay, A. K., Kelton, K. F. "Anomalous Thermal Expansion in Undercooled Liquid Ti39.5Zr39.5Ni21" 134th Annual TMS Meeting. Feb-2005